

---

## UNIT 17 UNCONSTRAINED AND CONSTRAINED OPTIMISATION

---

### Structure

- 17.0 Objectives
- 17.1 Introduction
- 17.2 Unconstrained Optimisation
  - 17.2.1 First Order Condition
  - 17.2.2 Second Order Condition
- 17.3 Some Economic Problems: Unconstrained Optimisation
- 17.4 Constrained Optimisation
  - 17.4.1 Equality Constrained Optimisation
  - 17.4.2 Inequality Constrained Optimisation
- 17.5 Kuhn- Tucker Theory
  - 17.5.1 The Kuhn-Tucker (KT) Theorem
- 17.6 Simple Economic Problems: Constrained Optimisation
- 17.7 Let Us Sum Up
- 17.8 Key Words
- 17.9 Answers or Hints to Check Your Progress Exercises
- 17.10 Exercises

---

### 17.0 OBJECTIVES

---

After going through this unit, you should be able to:

- determine the first order condition and second order condition with respect to unconstrained optimisation;
- solve unconstrained optimisation problems;
- apply the concept of unconstrained optimization on simple economic problems;
- solve the basic optimisation problems with equality as well as inequality constraint using Lagrange method; and
- use Kuhn-Tucker method while solving optimisation problems with inequality constraint.

---

### 17.1 INTRODUCTION

---

In the previous unit, we learnt about the extreme points of a function in terms of maxima and minima. We also understood the stationary point and inflexion point. Now, in this unit, our objective is to find and explain the necessary and sufficient conditions for any unconstrained optimisation problem as well as for constrained optimization problem. Optimisation in mathematical sense means either maximisation or, minimisation of different mathematical functions. Unconstrained optimisation deals with those problems whose domain is not

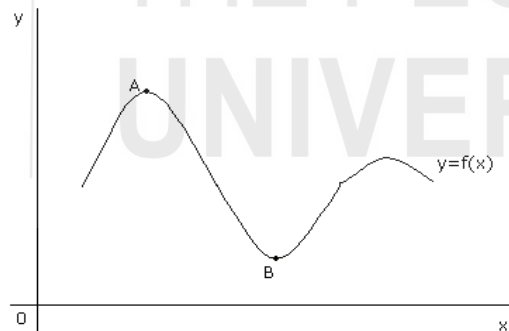
compressed by any constraint. We are going to use simple calculus to solve these kinds of optimisation problems. First, we find and discuss the necessary first order conditions for optimisation before moving on to the sufficient condition or, the second order condition. We will also work out some simple economic applications dealing with unconstrained and constrained optimisation.

## 17.2 UNCONSTRAINED OPTIMISATION

Under optimisation exercise we either maximise or, minimise some mathematical functions consisting of either one variable or, several variables. To optimise a mathematical function, we need to find the necessary first order conditions from which the optimum values of the variables can be found. But to find whether the value is maximum or, minimum we have to check the sufficient second order condition.

### 17.2.1 First Order Condition

Consider the function  $y = f(x)$ , which is shown in Fig 17.1. The function has the maximum point A and the minimum point B. Note that at these two points the value of  $y$  must be stationary. We can say that it is a necessary condition for an extremum of  $y$  that  $dy = 0$  as  $x$  varies. This condition is known as the differential version of the first order condition for an extremum and we have already discussed it in the preceding unit. As mentioned earlier, this condition is necessary but not sufficient for either a maximum or, a minimum. Therefore, the necessary first order condition for both maxima and minima are the same.



**Fig 17.1: First order condition for optimum**

We will see that the above condition is equivalent to the differential version of the first order condition  $\frac{dy}{dx} = 0$  or,  $f'(x) = 0$ . We note that when there is no change in  $x$  (i.e.,  $dx = 0$ ),  $dy$  will automatically be zero. But this, of course, is not what the first order condition is all about. What the first order condition requires is that  $dy$  be zero as  $x$  is varied, that is, as arbitrary (positive or, negative, but not zero) infinitesimal changes of  $x$  occur. In such a context, with  $dx \neq 0$ ,  $dy$  can be zero if and only if  $f'(x) = 0$ . Thus, the derivative condition  $f'(x) = 0$  and the differential condition  $dy = 0$  “for arbitrary nonzero values of  $dx$ ” are indeed equivalent.

Now, suppose  $y = f(x_1, x_2, \dots, x_n)$ . In this case, the first order condition for optimisation is given by the n-equation viz.,  $\frac{\partial y}{\partial x_1} = f^1(x) = 0$ ,  $\frac{\partial y}{\partial x_2} = f^2(x) = 0$ , ...,  $\frac{\partial y}{\partial x_n} = f^n(x) = 0$ . From these n equations, we can find  $x_1^*, x_2^*, \dots, x_n^*$ .

**Example 17.1:** Find the optimal values of  $y = x^3 - x$  in the domain  $-\infty < x < +\infty$ .

Ans. The necessary first order condition is  $\frac{dy}{dx} = 0$ . For the above equation we

have  $3x^2 - 1 = 0$  ----- (17.1)

From equation (17.1) we get values of x as either  $x^* = +\frac{1}{\sqrt{3}}$  or,  $x^* = -\frac{1}{\sqrt{3}}$ .

Putting these values of x, we get optimum y i.e.,  $y^*$ .

**Example 17.2:** Find the optimal values of  $z = y^2 - xy + x^2$  on the domain  $-\infty < x < +\infty$  and  $-\infty < y < +\infty$ .

Ans. The necessary first order condition gives  $\frac{\partial z}{\partial x} = 0$  and  $\frac{\partial z}{\partial y} = 0$ , from which

we can get  $-y + 2x = 0$  and  $2y - x = 0$ . We solve this and get  $x^* = 0$  and  $y^* = 0$ .

Finally putting these optimum values into the original equation we get  $z^* = 0$ .

## 17.2.2 Second Order Condition

For the time being, let us assume the function  $z = f(x, y)$  can give us two first order partial derivatives,  $f_x = \frac{\partial z}{\partial x}$  and  $f_y = \frac{\partial z}{\partial y}$ . Since  $f_x$  is itself a function of x (as well as y), we can measure the rate of change of  $f_x$  with respect to x, while y remains fixed, by a particular second order (or, second) partial derivative denoted by either  $f_{xx}$  or,  $\frac{\partial^2 z}{\partial x^2}$ .

$f_{xx} \equiv \frac{\partial}{\partial x}(f_x)$  Or,  $\frac{\partial^2 z}{\partial x^2} \equiv \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)$ . The notation  $f_{xx}$  has a double subscript signifying that the primitive function f has been differentiated partially with respect to x twice, whereas the notation  $\frac{\partial^2 z}{\partial x^2}$  resembles that of  $\frac{d^2 z}{dx^2}$  except for the use of the partial symbol. In a perfectly analogous manner, we can use the second partial derivative  $f_{yy} \equiv \frac{\partial}{\partial y}(f_y)$  or,  $\frac{\partial^2 z}{\partial y^2} \equiv \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)$  to denote the rate of change of  $f_y$  with respect to y, while x is held constant.

Recall however, that  $f_x$  is also a function of y. Hence, we can write two more partial derivatives:  $f_{xy} \equiv \frac{\partial^2 z}{\partial x \partial y} \equiv \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$  and  $f_{yx} \equiv \frac{\partial^2 z}{\partial y \partial x} \equiv \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ . These are called cross (or, mixed) partial derivatives because each measures the rate of

change of one first order partial derivative with respect to the “other” variable. We can also note that using total differential equation we get

$$d^2z \equiv d(dz) \equiv \frac{\partial(\partial z)}{\partial x} dx + \frac{\partial(\partial z)}{\partial y} dy.$$

Using the concept of  $d^2z$ , we can state the second order sufficient condition for a maximum of  $z = f(x, y)$  as follows:

$$d^2z < 0 \text{ for arbitrary values of } dx \text{ and } dy, \text{ not both zero} \quad \text{-----(a)}$$

The rationale behind this is very similar to that of the  $d^2z$  condition for the one variable case and it can be explained by means of a figure (Fig. 17.1) that depicts the bird's-eye view of a surface. Let point A on the surface – the point lying directly above the point  $(x_0, y_0)$  in the domain – satisfy the first order condition. Then the point A is a prospective candidate for a maximum. Whether it in fact qualifies depends on the surface configuration about A (Fig. 17.1). If an infinitesimal movement away from A in any direction along the surface invariably results in a decrease in  $z$ , i.e., if  $dz < 0$  for arbitrary values of  $dx$  and  $dy$ , not both zero then A is a pick of a dome. Given that  $dz = 0$  at point A, however, the condition  $dz < 0$  at point about A amounts to the stipulation that  $dz$  is decreasing, that is,  $d^2z \equiv d(dz) < 0$ , for arbitrary values of  $dx$  and  $dy$ , not both zero. Thus (a) constitutes a sufficient condition for identifying a stationary value as a maximum of  $z$ .

Analogous reasoning would show that a counterpart second order sufficient condition for identifying value as a minimum of  $z = f(x, y)$  is as follows:

$$d^2z > 0 \text{ for arbitrary values of } dx \text{ and } dy, \text{ not both zero} \quad \text{----- (b)}$$

The reason why (a) and (b) are sufficient conditions but not necessary conditions, is that it is again possible for  $d^2z$  to take a zero value at a maximum or, a minimum. For this reason, second order necessary conditions must be stated with weak inequalities as follows:

For *maximum* of  $z$ :  $d^2z \leq 0$  for arbitrary values of  $dx$  and  $dy$ , not both zero.

For *minimum* of  $z$ :  $d^2z \geq 0$  for arbitrary values of  $dx$  and  $dy$ , not both zero----(c)

In the following, however, we pay more attention to the second order sufficient conditions.

For operational convenience, second order differential conditions can be translated into equivalent conditions on second order derivatives. In the two-variable case, this would entail restrictions on the signs of the second order partial derivatives  $f_{xx}$ ,  $f_{yy}$  and  $f_{xy}$ . The actual translation would require knowledge of quadratic forms. But we may first introduce the main results here: For any values of  $dx$  and  $dy$ , not both zero, we have

$$d^2z < 0 \text{ if and only if } f_{xx} < 0, f_{yy} < 0, \text{ and } f_{xx}f_{yy} > f_{xy}^2$$

$$d^2z > 0 \text{ if and only if } f_{xx} > 0, f_{yy} > 0, \text{ and } f_{xx}f_{yy} > f_{xy}^2.$$

Note that the sign of  $d^2z$  hinges not only on  $f_{xx}$  and  $f_{yy}$ , which have to do with the surface configuration around point A (Fig. 17.1), but also on the cross partial derivative  $f_{xy}$ . The role played by this latter partial derivative is to ensure that the surface in question will yield (two-dimensional) cross sections with the same type of configuration in all possible directions. The above result, together with the first order condition, enables us to construct Table 17.1.

**Table 17. 1: Conditions for Relative Extremum:  $z=f(x, y)$**

| Condition                         | Maximum                                               | Minimum                                               |
|-----------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| First order necessary condition   | $f_x = f_y = 0$                                       | $f_x = f_y = 0$                                       |
| Second order sufficient condition | $f_{xx}, f_{yy} < 0$<br>and $f_{xx}f_{yy} > f_{xy}^2$ | $f_{xx}, f_{yy} > 0$<br>and $f_{xx}f_{yy} > f_{xy}^2$ |

We should understand that all the second order partial derivatives therein are to be evaluated at the stationary point where  $f_x = f_y = 0$ . It should also be stressed that the **second order sufficient condition is not necessary for an extremum**. If a stationary value is characterised by  $f_{xx}f_{yy} = f_{xy}^2$  then that stationary value may nevertheless turn out to be an extremum. On the other hand, in the case of **another type of violation**, with a stationary point characterised by  $f_{xx}f_{yy} < f_{xy}^2$ , we can identify that point as a saddle point, because the sign of  $d^2z$  will in that case be indefinite (positive for the same values of  $dx$  and  $dy$ , but negative for others).

For  $n$ -variable case we can generalise the above second order condition. The satisfaction of the first order condition earmarks certain values of  $z$  as the stationary values of the objective function. If at a stationary value of  $z$  we find that  $d^2z$  is positive definite, this will suffice to establish that value of  $z$  as a minimum. Analogously, the negative definiteness of  $d^2z$  is a sufficient condition for the stationary value to be maximum.

When there are  $n$  choice variables, the objective function may be expressed as  $z = f(x_1, x_2, \dots, x_n)$ .

The total differential will then be  $dz = f_1dx_1 + f_2dx_2 + \dots + f_ndx_n$  so that the necessary condition for extremum ( $dz = 0$  for arbitrary  $dx_i$ ) means that all the  $n$  first order partial derivatives are required to be zero.

The second order differential  $d^2z$  will again be a quadratic form, which can be expressed as an  $n \times n$  array. The coefficients of that array, properly arranged, will now give the (symmetric) Hessian matrix

$$|H| = \begin{vmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{vmatrix}$$

with principal minors  $|H_1| = f_{11}$ ,  $|H_2| = \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}$ , ...,  $|H_n| = |H|$ . The second order sufficient condition for extremum is, as before, that all the  $n$  principal minors be positive (for a minimum in  $z$ ) and that they duly alternate in sign (for a maximum in  $z$ ), the first one being negative.

In summary, then – if we concentrate on the determinantal test – we have the criteria as listed in Table 17.2, which is valid for an objective function of any number of choice variables. As special cases, we can have  $n = 1$  or,  $n = 2$ . When  $n = 1$ , the objective function is  $z = f(x)$ , and the condition for maximization,  $f_1 = 0$  and  $|H_1| < 0$ , reduce to  $f'(x) = 0$  and  $f''(x) < 0$ , exactly as we learned above. Similarly, when  $n = 2$ , the objective function is  $z = f(x_1, x_2)$ , so that the first order condition for maximum is  $f_1 = f_2 = 0$ , whereas the second order sufficient condition becomes  $f_{11} < 0$  and  $\begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix} = f_{11}f_{22} - f_{12}^2 > 0$ , which is merely a restatement of the information given above.

**Table 17.2: Determinantal Test for Relative Optimum:**  $z = f(x_1, x_2, \dots, x_n)$

| Condition                         | Maximum                                          | Minimum                                   |
|-----------------------------------|--------------------------------------------------|-------------------------------------------|
| First order necessary condition   | $f_1 = f_2 = \dots = f_n$                        | $f_1 = f_2 = \dots = f_n$                 |
| Second order sufficient condition | $(-1)^i  H_i  > 0 \quad \forall i = 1, \dots, n$ | $ H_i  > 0 \quad \forall i = 1, \dots, n$ |

**Example 17.3:** Find the extreme value(s) of  $z = x^2 + xy + 2y^2 + 3$  and determine whether they are maximum or, minimum.

Ans. The relevant first order conditions are the following:

$f_x = 2x + y = 0$  and  $f_y = x + 4y = 0$ . Solving these two equations we get  $x^* = 0$  and  $y^* = 0$ . Note that  $f_{xx} = 2 > 0$  and  $f_{yy} = 4 > 0$  and as  $f_{xy} = 1$  we have  $f_{xx}f_{yy} > f_{xy}^2$ . So at  $x^* = 0$  and  $y^* = 0$  the function  $z$  is minimized.

**Example 17.4:** Find the extreme value(s) of  $z = -x^2 + xy - y^2 + 2x + y$  and determine whether they are maximum or, minimum.

Ans. The relevant first order conditions are the following:

$f_x = -2x + y + 2 = 0$  and  $f_y = x - 2y + 1 = 0$ . Solving these two equations we get  $x^* = 5/3$  and  $y^* = 4/3$ . Now note that  $f_{xx} = -2 < 0$  and  $f_{yy} = -2 < 0$  and as  $f_{xy} = 1$  we have  $f_{xx}f_{yy} > f_{xy}^2$ . So, at  $x^* = 5/3$  and  $y^* = 4/3$  the function  $z$  is maximised.

## Check Your Progress 1

- 1) Find the extreme value(s) of  $z = -x^2 + xy - y^2 + x + 5y$  and determine whether they are maximum or minimum.

.....  
 .....  
 .....  
 .....

- 2) Find the extreme value(s) of  $y = x^2 - 6$  and determine whether they are maximum or minimum.

.....  
 .....  
 .....  
 .....

- 3) Find the extreme value(s) of  $y = x^3 - 2x^2 + x - 6$  and determine whether they are maximum or minimum.

.....  
 .....  
 .....  
 .....

- 4) Find the extreme value(s) of  $z = x^2 - 2x - y^2$  and determine whether they are maximum or minimum.

.....  
 .....  
 .....

---

## 17.3 SOME ECONOMIC PROBLEMS: UNCONSTRAINED OPTIMISATION

---

In this section, we will give some economic applications concerning unconstrained optimisation. Our first example is concerning the multi-product firm that takes the prices as given i.e., it produces under competitive structure. The second example is also regarding multi-product firm but here the firm acts as a monopolist where it can choose how much to produce and at what price. In both the cases the firm's objective is to maximise its own profit where profit is defined as net revenue minus net cost. These objectives we take for simplicity i.e., we can think of some other objective of the firm viz., revenue maximisation.

**Example 17.5:** This example is regarding the problem of a multi-product firm. Let us first postulate a two-product firm under circumstances of pure competition. Since, with pure competition the prices of both commodities must be taken as exogenous, these will be given by  $p_1 = 5$  and  $p_2 = 3$ , respectively. Accordingly, the firm's revenue function will be  $R = p_1q_1 + p_2q_2 = 5q_1 + 3q_2$ , where  $q_i$  represents the output level of the  $i^{\text{th}}$  product per unit of time. The firm's cost function is assumed to be  $C = 2q_1^2 + 2q_2^2 + q_1q_2$ .

Note that  $\frac{\partial C}{\partial q_1} = 4q_1 + q_2$  (the marginal cost of the first product) is a function not only of  $q_1$  but also of  $q_2$ . Similarly, the marginal cost of the second product also depends, in part, on the output level of the first product. Thus, according to the assumed cost function, the two commodities are seen to be technically related in production.

The profit function of this hypothetical firm can now be written readily as  $\pi = R - C = 5q_1 + 3q_2 - 2q_1^2 - 2q_2^2 - q_1q_2$ , a function of two choice variables ( $q_1$  and  $q_2$ ) and two price parameters. It is our task to find the levels of  $q_1$  and  $q_2$ , which in combination, will maximise  $\pi$ . For this purpose, we find the first order partial derivatives of the profit function:

$$\pi_1 \left( \equiv \frac{\partial \pi}{\partial q_1} \right) = 5 - 4q_1 - q_2 \text{ and } \pi_2 \left( \equiv \frac{\partial \pi}{\partial q_2} \right) = 3 - 4q_2 - q_1 \quad \dots (a)$$

Setting these both equal to zero, to satisfy the necessary condition for maximum, we get the two simultaneous equations  $4q_1 + q_2 = 5$  and  $q_1 + 4q_2 = 3$ , which yield the unique solution  $q_1^* = \frac{4p_1 - p_2}{15} = \frac{17}{15}$  and  $q_2^* = \frac{4p_2 - p_1}{15} = \frac{7}{15}$ .

Thus, the maximum profit that the firm will get is  $\pi^* = 5.51$  (approx).

To be sure that this does represent a maximum profit, let us check the second order condition. The second order partial derivatives, obtainable by partial differentiation of equation (a), give us the following Hessian matrix:

$$|H| = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ -1 & -4 \end{bmatrix}$$

Since  $|H_1| = -4 < 0$  and  $|H_2| = 15 > 0$ , the Hessian matrix (or,  $d^2z$ ) is negative definite, and the solution does maximise the profit. In fact, since the signs of the principal minors do not depend on where they are evaluated,  $d^2z$  is in this case everywhere negative definite. Thus, the objective function must be strictly concave, and the maximum profit found above is a unique absolute maximum.

**Example 17.6:** This example is again regarding a multi-product firm but with the modification that in earlier case the firm faced a given price i.e., multi-product firm under competitive structure. In this example the multi-product firm is a monopoly in both the products i.e., the firm can decide how much quantity to be produced and at the same time it can choose any price for its product.

Now by virtue of this new market-structure assumption, the revenue function must be modified to reflect the fact that the prices of the two products will now vary with their output levels (which are assumed to be identical with their sales levels, no inventory accumulation being contemplated in the model). The exact way prices will vary with output levels is, of course, to be found in the demand functions for the firm's two products.

Suppose that the demand facing the monopolist firm, are as follows:

$$q_1 = 40 - 2p_1 + p_2 \text{ and } q_2 = 15 + p_1 - p_2 \quad \dots\dots\dots (17.2)$$

These equations reveal that the two commodities are related in consumption; specifically, they are substitute goods, because an increase in the price of one will raise the demand for the other. As given, equation (a) expresses the quantities demanded  $q_1$  and  $q_2$  as functions of prices, but for our present purposes, it will be more convenient to have prices  $p_1$  and  $p_2$  expressed in terms of the sales volumes  $q_1$  and  $q_2$ , that is, to have average revenue functions for the two products. Since equation (17.2) can be rewritten as

$-2p_1 + p_2 = q_1 - 40$  and  $p_1 - p_2 = q_2 - 15$ , we may (considering  $q_1$  and  $q_2$  as parameters) apply Cramer's rule to solve for  $p_1$  and  $p_2$  as follows:

$$p_1 = 55 - q_1 - q_2 \text{ and } p_2 = 70 - q_1 - 2q_2 \quad \dots\dots\dots (17.3)$$

These constitute the desired average-revenue functions, since  $p_1 \equiv AR_1$  and  $p_2 \equiv AR_2$ . Consequently, the firm's total-revenue function can be written as:

$$\begin{aligned} R &= p_1 q_1 + p_2 q_2 \\ &= (55 - q_1 - q_2)q_1 + (70 - q_1 - 2q_2)q_2 \\ &= 55q_1 + 70q_2 - 2q_1 q_2 - q_1^2 - 2q_2^2 \end{aligned}$$

If we again assume the total cost function to be  $C = q_1^2 + q_1 q_2 + q_2^2$ , then the profit function will be:

$$\begin{aligned} \pi &= R - C \\ &= 55q_1 + 70q_2 - 3q_1 q_2 - 2q_1^2 - 3q_2^2 \quad \dots\dots\dots (17.4) \end{aligned}$$

which is an objective function with two choice variables. Once the profit maximizing output levels  $q_1^*$  and  $q_2^*$  are found, however, the optimal prices  $p_1^*$  and  $p_2^*$  are easy enough to find from equation (17.3).

The objective function yields the following first order and second order partial derivatives:

$$\begin{aligned} \pi_1 &= 55 - 3q_2 - 4q_1 & \pi_2 &= 70 - 3q_1 - 6q_2 \\ \pi_{11} &= -4 & \pi_{12} = \pi_{21} &= -3 & \pi_{22} &= -6 \end{aligned}$$

To satisfy the first order condition for a maximum of  $\pi$ , we must have  $\pi_1 = \pi_2 = 0$ ; that is  $4q_1 + 3q_2 = 55$  and  $3q_1 + 6q_2 = 70$ . Thus, the solution output levels (per unit of time) are

$$(q_1^*, q_2^*) = (8, 7\frac{2}{3}).$$

Upon substitution of this result into equation (17.3) and (17.4), respectively, we find that

$$p_1^* = 39\frac{1}{3} \quad p_2^* = 46\frac{2}{3} \quad \text{and} \quad \pi^* = 488\frac{1}{3} \quad (\text{per unit of time})$$

In so much as the Hessian is  $\begin{bmatrix} -4 & -3 \\ -3 & -6 \end{bmatrix}$ , we have  $|H_1| = -4 < 0$  and  $|H_2| = 15 > 0$ ,

so that the value of  $\pi^*$  does represent the maximum profit. Here, the signs of the principle minors are again independent of where they are evaluated. Thus, the Hessian matrix is everywhere negative definite, implying that the objective function is strictly concave and that it has a unique absolute maximum.

---

## 17.4 CONSTRAINED OPTIMISATION

---

When an economic project such as the production of a specified level of output is to be carried out, there are normally several alternative ways of accomplishing it. One (or more) of these alternatives will, however, be more desirable than others from the standpoint of a producer. Need, therefore, arises to choose the best alternative available without compromising the basic objective of the project. Essentially, the constraint optimisation helps arrive the decision.

The most common criterion of choice among alternatives in economics is the goal of maximising something (such as maximising a firm's profit, a consumer's utility, or the rate of growth of firm or of a country's economy) or, of minimising something (such as minimising the cost of producing a given output). Economically, we may categorize such maximisation and minimisation problems under the general heading of optimisation, meaning "the quest for the best". From a purely mathematical point of view, however, the terms maximum and minimum do not carry with them any connotation of optimality. Therefore, the collective term for maximum and minimum, as mathematical concepts, is the more matter-of-fact designation extremum, meaning an extreme value.

In formulating an optimisation problem, the first order of business is to delineate an objective function in which the dependent variable represents the object to maximisation or minimisation and in which the set of independent variables indicates the objects whose magnitudes the economic unit in question can pick and choose, with a view to optimising. We shall, therefore, refer to the independent variables as choice variables. They can also be called decision variables, or policy variables. The essence of the optimisation process is simply